

Constraint Reaction Forces and Lagrange Multipliers in Multi-Body Systems

Wenli Yao^{1, a}, Yongsheng Ren^{2, b}

¹ Science College, Qingdao Technological University, Qingdao, P.R.China

² College of Mechanical and Electronic Engineering, Shandong University of Science and Technology, Qingdao, P.R.China

^aywenli1969@sina.com, ^brenys@sdust.edu.cn

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Abstract. The study performs the relation between the Lagrange multipliers and the constraint reaction forces in the multi-body systems. It helps to establish a simple and effective method to deal with the non-smooth factors including friction, contact and collision in multi-body systems. First, a general expression calculating the ideal constraint reaction forces is described by a form of complete Cartesian reference coordinate in a discrete dynamical system. Then, an explicit expression is given in the form of independent general generalized coordinates for a multi-body system. A condition under which the constraint reaction forces can be one-to-one corresponding to Lagrange multipliers is discussed. An example is used to demonstrate the process to realize the condition and the effectiveness of this method.

Introduction

Dynamics of multi-body systems is the important base of theory for design and control of the complex mechanical systems, such as vehicles, manufacturing equipment, robots and aerospace. The dynamical theory of smooth multi-body systems has been mature[1,2]. When the non-smooth factors including friction, contact, collision cannot be ignored in the multibody systems, some difficulties arise in the use of the usual methods[3]. When dry friction in the joints is considered, the friction force needs to be expressed as the function of the normal reaction forces. The normal constraint reaction forces can be obtained by D'Alembert principle[4], but this method is difficult to be combined with dynamics of multi-body systems. In the two traditional methods including Lagrange and Cartesian method for multi-body systems, the normal constraint forces can be introduced into the dynamical equations in the form of the Lagrange multipliers. Some authors usually equal the constraint reaction forces to the Lagrange multipliers. Edward gave the proof on the relation between the two physical variables for a discrete system by the complete Cartesian coordinates[5]. An approach to solve constraint forces of multi-body system was presented by means of generalized Cartesian coordinate and partial approach of constraint equations [6]. In this paper, an explicit expression of constraint reaction forces is derived in the form of the Lagrange multipliers for more general Lagrange systems. With the help of the above expressions, the normal constraint reaction forces in joints are related to the Lagrange multipliers one-to-one.

Expression of the Constraint Reaction Force in the form of Complete Cartesian Coordinates

Let us consider a discrete system consisting of N particle. The vector $\mathbf{r} = [x_1, \dots, x_{3N}]^T$ specifies the position of the system, using the Cartesian coordinates $(x_{3i-2}, x_{3i-1}, x_{3i})$ to represent the i th particle.

The kinematical constraints on the system can be written in the form

$$\boldsymbol{\varphi}(r, t) = [\varphi_1(r, t), \dots, \varphi_m(r, t)]^T = \mathbf{0} \quad (1)$$

A virtual displacement $\delta \mathbf{r}$ is kinematically admissible if

$$\boldsymbol{\varphi}_r \delta \mathbf{r} = \mathbf{0} \quad (2)$$

The ideal constraint force $\mathbf{F}^C$ does not work for all $\delta \mathbf{r}$ satisfying Eq.2, such that

$$\delta \mathbf{r}^T \mathbf{F}^C = \delta W^C = 0 \quad (3)$$

According to Lagrange multiplier theorem, there exists a Lagrange multiplier vector $\boldsymbol{\lambda} \in R^m$ such that

$$\delta \mathbf{r}^T \mathbf{F}^C + \delta \mathbf{r}^T \boldsymbol{\varphi}_r^T \boldsymbol{\lambda} = 0 \quad (4)$$

Hence the explicit expression of the constraint force can be written as

$$\mathbf{F}^C = -\boldsymbol{\varphi}_r^T \boldsymbol{\lambda} \quad (5)$$

If $\boldsymbol{\varphi}_r$ has full row rank, the Lagrange multipliers uniquely characterize the constraint forces. It is sometimes stated in the literature on dynamics that Lagrange multipliers are constraint forces. As shown in the above, this is not correct.

Expression of the Constraint Reaction Force in a Generalized Coordinate Space

Consider the above discrete system, which has n degrees of freedom. Suppose the configuration of the system is specified by the generalized coordinates $q_1, q_2, \dots, q_l$ and assume that there are k independent equations of constraint. It is convenient to think of these coordinates and the constraint equation as the vectors in the following form, respectively:

$$\mathbf{q} = [q_1, q_2, \dots, q_l]^T, \quad \boldsymbol{\varphi}(q, t) = [\varphi_1(q, t), \dots, \varphi_k(q, t)]^T = \mathbf{0}$$

Hence the generalized coordinate variations $\delta \mathbf{q}$ must satisfy

$$\boldsymbol{\varphi}_q \delta \mathbf{q} = \mathbf{0} \quad (6)$$

Assume the constraint forces arise from contact in the joints. If the position vector of the contact points can be parameterized by the generalized coordinate vector $\mathbf{r}_C$, the relationship between generalized coordinate variations and the virtual displacements $\delta \mathbf{r}_C$ is

$$\delta \mathbf{r}_C^T = (\mathbf{r}_q \delta \mathbf{q})^T = \delta \mathbf{q}^T \mathbf{r}_q^T \quad (7)$$

According to the definition of ideal constraint reaction forces, we have

$$\delta \mathbf{r}_C^T \mathbf{F}^C = \delta \mathbf{q}^T \mathbf{r}_q^T \mathbf{F}^C = 0 \quad (8)$$

Lagrange multiplier theorem may be applied to Eq.8, which must hold for all $\delta \mathbf{r}_C$ that satisfy Eq.7. Thus there must exist a Lagrange multiplier vector such that

$$\delta \mathbf{q}^T \mathbf{r}_q^T \mathbf{F}^C + \delta \mathbf{q}^T \boldsymbol{\varphi}_q^T \boldsymbol{\lambda} = 0 \quad (9)$$

Then Eq.9 yields

$$\mathbf{r}_q^T \mathbf{F}^C = -\boldsymbol{\varphi}_q^T \boldsymbol{\lambda} \quad (10)$$

Multiply both sides of Eq.10 on the left by $\mathbf{r}_q$. If $\mathbf{r}_q$ has full row rank, $\mathbf{r}_q \mathbf{r}_q^T$ is nonsingular such that

$$\mathbf{F}^C = -(\mathbf{r}_q \mathbf{r}_q^T)^{-1} \mathbf{r}_q \boldsymbol{\varphi}_q^T \boldsymbol{\lambda} = -(\mathbf{r}_q^T)^{-1} \boldsymbol{\varphi}_q^T \boldsymbol{\lambda} \quad (11)$$

According to Eq.11, we have the following conclusion:

When $\mathbf{r}_q = -\boldsymbol{\varphi}_q$, i.e., the form of the constraint equations is the same as the position expressions of the contact points as the independent generalized coordinates, the normal constraint forces are related to the Lagrange multipliers one-to-one.

Example

A crank-slider system is shown in Fig.1. The moment M is applied on the crank AB . The mass of the crank AB is m , the mass of the slider as m . Neglecting the mass of the connecting rod BC , determine the normal reaction force on the slider.

We choose the angles θ and φ as the generalized coordinates, as shown in Fig.2.

The kinematic energy of the system is

$$T = \frac{1}{2} J_A \dot{\theta}^2 + \frac{1}{2} m(\dot{x}_C^2 + \dot{y}_C^2) = 2/3 mr^2 \dot{\theta}^2 + 1/2 ml^2 \dot{\varphi}^2 + mrl \sin(\varphi - \theta) \dot{\theta} \dot{\varphi} \quad (12)$$

The constraint equation is that

$$y_C = r \sin \theta - l \cos \varphi = \text{const.} \quad (13)$$

According to the first kind of Lagrange equations, we have the following dynamical equations of the system

$$\begin{bmatrix} \frac{4}{3} mr^2 & mrl \sin(\varphi - \theta) & -r \cos \theta \\ ml^2 & mrl \sin(\varphi - \theta) & -l \sin \varphi \\ r \cos \theta & l \sin \varphi & 0 \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\varphi} \\ \lambda \end{bmatrix} = \begin{bmatrix} \frac{3}{2} mgr \sin \theta + M - mrl \cos(\varphi - \theta) \dot{\varphi}^2 \\ mgl \sin \varphi + mrl \cos(\varphi - \theta) \dot{\theta}^2 \\ r \sin \theta \dot{\theta}^2 - l \cos \varphi \dot{\varphi}^2 \end{bmatrix} \quad (14)$$

Solving Eq.14, we can obtain the explicit expression of the multiplier λ , which is connected with the configuration, angular velocities, applied force and etc. According to the above conclusion, for the constraint equation in this example, we can obtain the normal constraint force of the slider corresponding to the multiplier λ one-to-one.

When the following parameters are chosen:

$$m = 1 \text{ kg}, r = 1 \text{ m}, l = \sqrt{3} \text{ m}, \theta = \pi/3, \dot{\theta} = 1 \text{ rad/s}, g = 9.8 \text{ m/s}^2, \varphi = \pi/3, \dot{\varphi} = -1/3 \text{ rad/s}$$

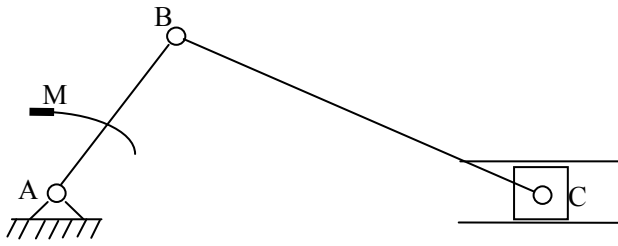


Fig.1 A crank-slider system

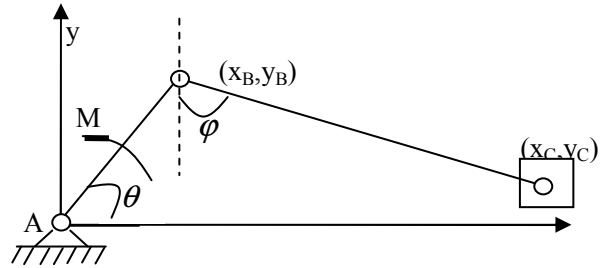


Fig.2 The generalized coordinates of the system

We have $F_{NC} = \lambda = 0.7698N$, which agrees with the one calculated by D'Alembert principle.

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Conclusions

The usual method calculating ideal constraint reaction forces is inconvenient to form a general computational program, which is the key factor for dynamics of the multi-body systems. This paper provides us a more simple and stylized method to deal with non-smooth dynamical problems in multi-body systems.

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